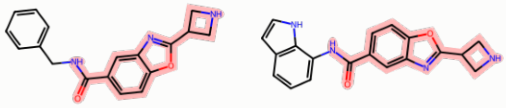
THREE CLOSEST PAIRS — each method's WORST case

all three libraries built at the SAME fixed 100-reaction budget · T = Tanimoto on the mode metric (Morgan r=3 / 2048)

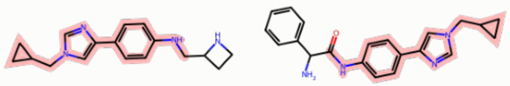
the shared core (MCS) is haloed in red — what stays black is what actually differs

hub-batching (ours) (82 molecules)

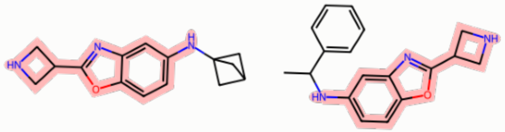
T 0.50 · 70% shared



T 0.50 · 81% shared

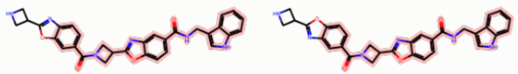


T 0.50 · 74% shared

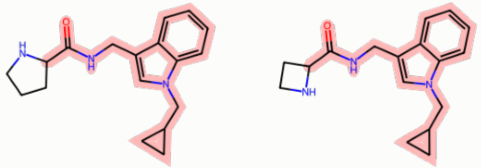


BC-Enum-SB (λ=1) (96 molecules)

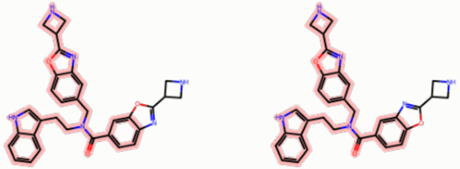
T 0.87 · 83% shared



T 0.86 · 86% shared

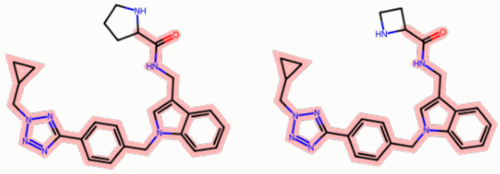


T 0.83 · 83% shared

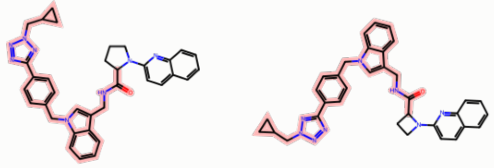


BC-SB (λ=1) (91 molecules)

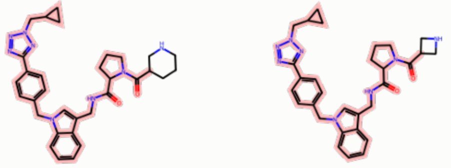
T 0.90 · 91% shared



T 0.89 · 70% shared



T 0.88 · 92% shared



The pairs the metric is really about: how redundant does each library get? A pair above 0.5 would be merged into ONE "distinct molecule" by the mode metric, so our worst case sits at the 0.50 ceiling the selection enforces, while the competitors tolerate 0.83-0.90.